

Project Title

Project subtitle (if applicable)

Your names | Group #X

2026-03-16

```
{r setup, include=FALSE} library(gridExtra) library(tidyverse) library(knitr)
library(cowplot) knitr::opts_chunk$set( echo = FALSE, include = TRUE, warning =
FALSE, message = FALSE, results = "markup", fig.width = 6, fig.height = 3.5, fig.align =
"center", out.width = "80%" )

{r load-data, echo=FALSE} ### Read in data CATF_data <-
read_csv("CATF_data.csv")

{r clean_data, include=FALSE}
```

Clean the data

E.g., remove unrealistic populations

```
#CATF_data |> arrange(population) |> View()
```

```
#CATF_data <- CATF_data |> filter(population > 100) ## Alternatively, we could  
convert them to the correct values for each country by ## seeing roughly what the  
population values for each country are and scaling up accordingly.
```

Section 1



Section 2

Aggregate CATF_data: one row per country

Section 3



Section 4

Each country has multiple rows (one per student). We collapse

Section 5

them into a single row using sensible summary functions.

Section 6

Section 7

General strategy:

Section 8

- numeric columns $\rightarrow$ median (robust to outliers / typos)

Section 9

- character columns $\rightarrow$ mode (most common value reported)

Section 10



Section 11

Feel free to swap `median()` for `mean()` or `mode()` on any

Section 12

variable if you think a different choice is more appropriate.

Section 13



Section 14

— helper function: statistical mode (most frequent value) —

Section 15

works for both character and numeric vectors

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```
stat_mode <- function(x) { x <- x[!is.na(x)] if (length(x) == 0) return(NA) ux <-  
unique(x) ux[which.max(tabulate(match(x, ux)))] }
```

Section 16

— identify column types —————

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```
numeric_cols <- CATF_data |> select(where(is.numeric)) |> names()
```

```
character_cols <- CATF_data |> select(where(is.character), -country) |> names()
```

Section 17

— aggregate —

```
CATF_clean <- CATF_data |> group_by(country) |> summarise( # For every numeric
column, take the median (ignoring NAs) across(all_of(numeric_cols), function(x){
median(x, na.rm = TRUE) })),

# For every character column, take the mode
across(all_of(character_cols), stat_mode),

# (optional) track how many student rows were combined
n_sources = n()

)
```


Section 18

— quick check —

```
#glimpse(CATF_clean)
```

Section 19

How many unique countries do we have?

How many unique countries do we have?

```
#nrow(CATF_clean)
```

Section 20



Section 21

CUSTOMISATION IDEAS

Section 22



Section 23

If you'd prefer to use the mean for some variables and the

Section 24

median for others, you can split them up like this:

Section 25

Section 26

```
CATF_clean <- CATF_data |>
```

Section 27

```
group_by(country) |>
```

Section 28

summarise(

Section 29

Variables where mean makes sense

Section 30

```
across(c(gdp, population),
```

Section 31

```
function(x){ mean(x, na.rm = TRUE) } ),
```


Section 32

Section 33

Variables where median is more robust

Section 34

```
across(c(poverty_rate, hdi, median_age),
```

Section 35

```
function(x){ median(x, na.rm = TRUE) } ),
```

Section 36

Section 37

Variables where mode makes sense (e.g. categorical-ish)

Section 38

```
across(c(renewable_target_solar, renewable_target_wind),
```

Section 39

`stat_mode),`

Section 40

Section 41

```
n_sources = n()
```

Section 42

)

Section 43



Introduction

Overall Theme: [State your unifying theme]

Research Goal: [What are you investigating?]

Target Population: [Who/what is being studied?]

Audience: [Who is this for?]

Data Summary

Key Variables:

Variable 1: [Description]

Variable 2: [Description]

Data Wrangling:

[Any transformations, new variables created]

Section 44

Here is an example of a nicely formatted table, using the kable function

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```
iris |> group_by(Species) |> summarise( Count = n(), Avg Sepal Length =  
round(mean(Sepal.Length), 1), Avg Petal Length = round(mean(Petal.Length), 1) ) |>  
kable(caption = "Comparison of Iris Species")
```

Insights for Q1

Question #2: [Name - Question Statement]

[Repeat structure from Q1]

Q2: Variables & Approach

Q2: Visualization

Q2: Summary Table

Question #3: [Name - Question Statement]

Q3: Variables & Approach